



**SURE ProEd**

**MAJOR PROJECT REPORT  
(AutoCAD)**

**Title:** Train Using AutoCAD

**SUBMITTED BY**

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**Batch:** G13 - AutoCAD

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## Abstract

This project focuses on the preparation of a complete 2D AutoCAD drawing of a rail-based vehicle using standard engineering drawing practices. The drawing includes multiple orthographic views, an isometric view, and a standardized title block. The objective of the project is to demonstrate tools in AutoCAD, drafting accuracy, scaling, layering, and adherence to industrial drawing standards. The final output represents a professional technical drawing suitable for academic evaluation and introductory industrial applications.

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## 1. Introduction

Computer Aided Design (CAD) plays a vital role in modern mechanical and transportation engineering. AutoCAD is one of the most widely used drafting software tools for preparing precise technical drawings. Rail vehicles such as locomotives and maintenance trains require accurate drawings for design, analysis, manufacturing, and maintenance.

In this major project, a rail vehicle model is drafted in AutoCAD using different views including front view, side view, top view, and isometric view. The project emphasizes correct proportions, and neat presentation, which are essential skills for mechanical and production engineers.

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## 2. Objectives of the Project

The main objectives of this project are:

- To understand and apply AutoCAD drafting tools effectively
  - To create accurate orthographic projections of a rail vehicle
  - To prepare an isometric view for better visualization
  - To apply proper dimensioning and annotation techniques
  - To follow standard engineering drawing conventions
  - To develop a professional layout with title block and scale
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## 3. Software and Tools Used

- **Software:** AutoCAD
- **Drawing Type:** 2D Drafting
- **Units Used:** Millimeters (mm)
- **Drafting Environment:** Model Space and Layout Space

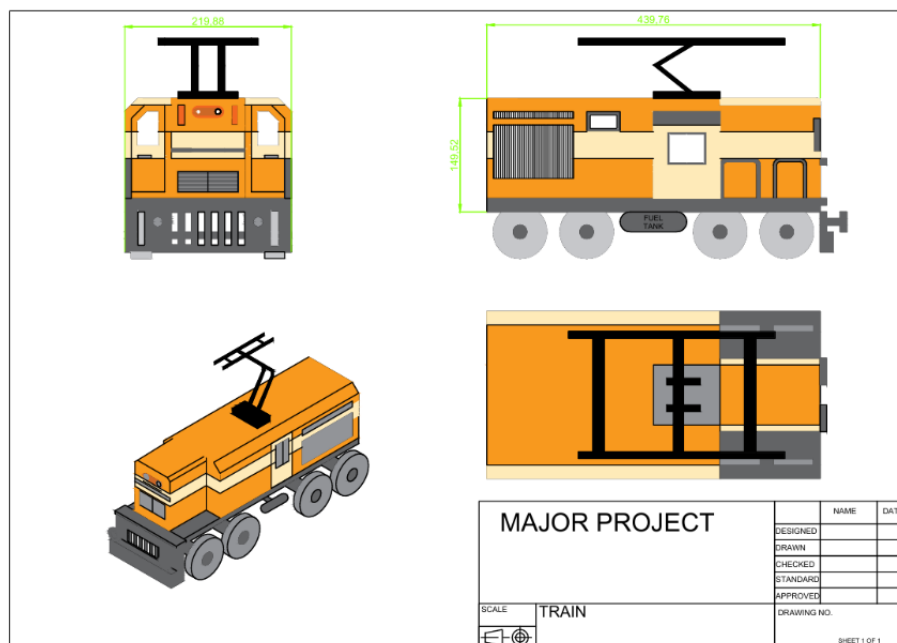
## AutoCAD Commands Used

- Line
- Polyline
- Rectangle
- Circle
- Arc
- Offset
- Trim and Extend
- Mirror
- Copy
- Move
- Hatch
- Dimension (Linear, Aligned)
- Layer Management
- Text

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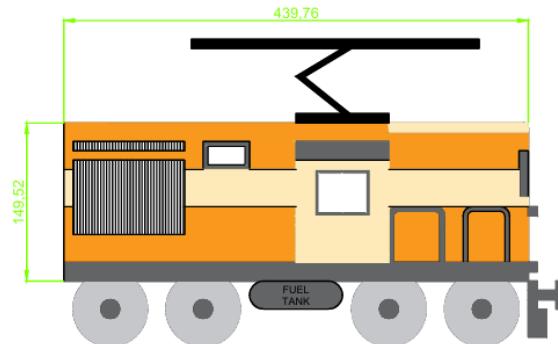
## 4. Description of the Drawing

The AutoCAD drawing represents a rail vehicle designed for electric traction or maintenance purposes. The drawing includes the following views:



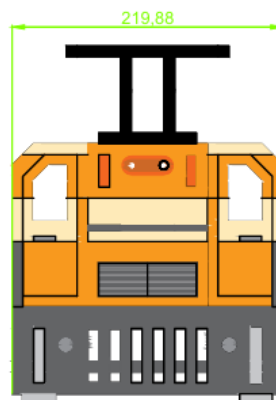
#### 4.1 Front View

The front view shows the overall width and height of the vehicle. Key features such as headlights, buffers, front grille, and pantograph base are visible. This view helps in understanding the frontal profile and symmetry of the train.



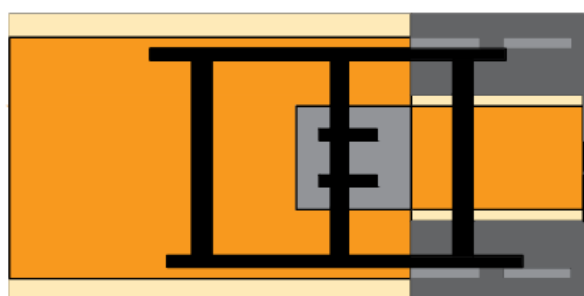
#### 4.2 Side View

The side view represents the complete length of the rail vehicle along with wheel arrangement, doors, windows, ventilation grills, and pantograph structure. This view is crucial for understanding the longitudinal layout and component placement.



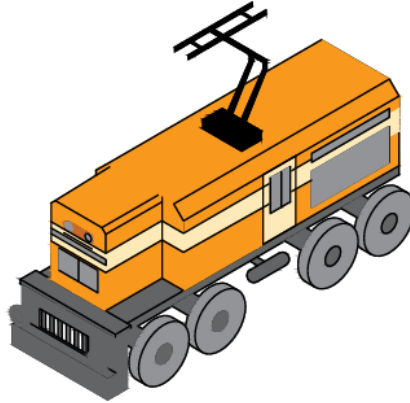
#### 4.3 Top View

The top view illustrates the roof arrangement, pantograph position, equipment layout, and roof clearances. It provides a clear idea of the spatial arrangement of components when viewed from above.



#### 4.4 Isometric View

The isometric view gives a three-dimensional appearance of the train, improving visualization and presentation quality. It helps viewers understand the shape and structure of the vehicle more easily compared to 2D views alone.



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#### 5. Dimensioning and Scale

- Proper dimensions are provided for length, width, and height
- Dimensions are placed clearly without overlapping
- Standard dimension styles are used
- The drawing is prepared to scale, ensuring proportional accuracy

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#### 6. Layers and Colour Coding

Different layers are used to improve clarity and organization of the drawing:

- Body outline layer
- Wheel and track layer
- Dimensions layer
- Text and annotations layer
- Hatch and colour fill layer

Colour coding enhances visual distinction between different components and improves readability.

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## **7. Title Block and Drawing Details**

The drawing includes a standard title block containing:

- Project Title: MAJOR PROJECT
- Drawing Name: TRAIN
- Scale
- Drawn by
- Checked by
- Approved by
- Date

This ensures the drawing follows professional drafting standards.

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## **8. Applications of the Project**

- Educational demonstration of AutoCAD skills
  - Basic reference for rail vehicle layout design
  - Useful for understanding drafting standards in mechanical engineering
  - Can be extended for 3D modelling and simulation
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## **9. Advantages of Using AutoCAD**

- High accuracy and precision
  - Easy modification and editing
  - Professional-quality drawings
  - Time saving compared to manual drafting
  - Industry-accepted software
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## **10. Limitations of the Project**

- The drawing is a conceptual representation
- Internal components are not detailed
- Structural and stress analysis is not included
- Electrical and mechanical systems are not modelled

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## **11. Conclusion**

The AutoCAD major project successfully demonstrates the drafting of a rail vehicle using standard engineering drawing principles. The project enhanced understanding of orthographic projections, isometric views, dimensioning, and drawing presentation. This work reflects the practical application of CAD knowledge and provides a strong foundation for advanced design and modelling projects.

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## **12. References**

1. AutoCAD
2. Search Engine (Image from Google)